

House Price Prediction Using Machine Learning

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various housing features such as area, number of bedrooms, bathrooms, stories, parking facilities, and other amenities. The project aimed to identify the factors that most influence house prices and evaluate the performance of different regression models.

Dataset

The Housing Prices Dataset was obtained from Kaggle. The dataset contains information about residential properties including structural features, amenities, and selling prices. The target variable for prediction was the house price.

Data Preparation

The dataset was explored and checked for missing values and duplicate records. No duplicate rows were found. Categorical features such as main road access, guest room availability, air conditioning, and furnishing status were converted into numerical format using one-hot encoding to make them suitable for machine learning algorithms.

Model Building

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio.

Results

MODEL	MAE	RMSE	R ² SCORE
Linear regression	970043	1324507	0.653
Random forest	1021546	1400566	0.612

Linear Regression performed better than Random Forest on this dataset, achieving the highest R² score and lower prediction errors.

Key Insights

- Property area is one of the strongest factors influencing house prices.
- The number of bathrooms and premium amenities significantly affect pricing.
- Houses located in preferred areas generally have higher market value.
- Linear Regression explained approximately 65% of the variation in house prices.

Recommendation

Real estate businesses should focus on property size, number of bathrooms, and premium amenities when estimating property values and designing pricing strategies. These features have a significant impact on overall house prices.

Conclusion

The project successfully demonstrated the use of machine learning techniques for house price prediction. After data preprocessing, model training, evaluation, and visualization, Linear Regression emerged as the best-performing model for this dataset.